

Summations and geometric progressions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Summations

Definition 1: summation

Let $a_1, a_2, \dots, a_n$ be n real numbers. Their sum

$$a_1 + a_2 + \dots + a_n$$

can be written in compact form with the summation symbol:

$$\sum_{k=1}^n a_k$$

which is read: “sum for k from 1 to n of a_k ”. The symbol k is called the summation index.

- The summation symbol is therefore just a shorthand, which is nevertheless very useful when the terms a_k are defined explicitly as a function of the index k .

Example 1: summations

$$\sum_{k=1}^{10} \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}$$

$$\sum_{k=3}^n k^2 = 3^2 + 4^2 + 5^2 + \dots + n^2$$

- The summation index is a **dummy index**. This means that if we replace k with i , j or any other index (in all its occurrences), the value of the summation does not change.

Example 2: dummy index

We have:

$$\sum_{k=1}^n k^2 = \sum_{i=1}^n i^2$$

since both symbols denote the sum of the squares of the first n natural numbers (without zero).

On the other hand, we have:

$$\sum_{k=1}^n k^2 \neq \sum_{k=1}^m k^2$$

since the two symbols denote the sum of, respectively, the first n or the first m squares (if $n \neq m$ the result will be different).

1.1 Main properties of summations

Remark 1

Given $c \in \mathbb{R}$, we have:

$$\sum_{k=1}^n (c \cdot a_k) = c \sum_{k=1}^n a_k \quad (\text{product by a constant}) \quad (1)$$

$$\sum_{k=1}^n c = c \cdot n \quad (\text{summation with a constant term}) \quad (2)$$

Proof. First property (1). By the distributive property we have:

$$\underbrace{c a_1 + c a_2 + \cdots + c a_n}_{=\sum_{k=1}^n (c \cdot a_k)} = \underbrace{c (a_1 + a_2 + \cdots + a_n)}_{=c \sum_{k=1}^n a_k}$$

Second property (2):

$$\underbrace{c + c + \cdots + c}_{=\sum_{k=1}^n c \quad \text{that is, } c \text{ added } n \text{ times}} = c n$$

□

Remark 2

Given two summations $\sum_{k=1}^n a_k$ and $\sum_{k=1}^n b_k$, we have:

$$\sum_{k=1}^n a_k + \sum_{k=1}^n b_k = \sum_{k=1}^n (a_k + b_k) \quad (3)$$

Proof. We have:

$$\underbrace{a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n}_{=\sum_{k=1}^n a_k + \sum_{k=1}^n b_k} = \underbrace{a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n}_{=\sum_{k=1}^n (a_k + b_k)}$$

□

Remark 3

Given two natural numbers $n, m \in \mathbb{N}$, we have:

$$\sum_{k=1}^{n+m} a_k = \sum_{k=1}^n a_k + \sum_{k=n+1}^{n+m} a_k \quad (\text{splitting}) \quad (4)$$

$$\sum_{k=1}^n a_k = \sum_{k=1+m}^{n+m} a_{k-m} = \sum_{k=1-m}^{n-m} a_{k+m} \quad (\text{index shift}) \quad (5)$$

$$\sum_{k=1}^n a_k = \sum_{k=1}^n a_{n-k+1} = \sum_{k=0}^{n-1} a_{n-k} \quad (\text{index reflection}) \quad (6)$$

Proof. The three properties are simply different ways of writing and/or ordering the terms of the summations. $\square$

1.2 Some important summations**Remark 4: sum of the first n natural numbers (without zero)**

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Proof.

$$\begin{aligned} \sum_{k=1}^n k &= \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n k \right) = \frac{1}{2} \left(\sum_{k=1}^n k + \sum_{k=1}^n (n - k + 1) \right) \\ &= \frac{1}{2} \sum_{k=1}^n (k + n - k + 1) = \frac{1}{2} \sum_{k=1}^n (n + 1) = \frac{n(n+1)}{2} \end{aligned}$$

$\square$

Remark 5: sum of the first n odd numbers

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n (2k-1) = n^2 \quad \text{or equivalently} \quad \sum_{k=0}^{n-1} (2k+1) = n^2$$

Proof. Using the properties of summations we have:

$$\begin{aligned} \sum_{k=1}^n (2k-1) &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \left(\frac{n(n+1)}{2} \right) - n = n^2 + n - n = n^2 \\ \\ \sum_{k=0}^{n-1} (2k+1) &= 2 \sum_{k=0}^{n-1} k + \sum_{k=0}^{n-1} 1 \\ &= 2 \sum_{k=1}^n (k-1) + n = 2 \left(\sum_{k=1}^n k - \sum_{k=1}^n 1 \right) + n \\ &= 2 \left(\frac{n(n+1)}{2} - n \right) + n = n^2 + n - 2n + n = n^2 \end{aligned}$$

□

Remark 6: sum of the first n even numbers (without zero)

For every natural number $n \geq 1$, we have:

$$\sum_{k=1}^n 2k = n(n+1)$$

Proof.

$$\sum_{k=1}^n 2k = 2 \sum_{k=1}^n k = 2 \left(\frac{n(n+1)}{2} \right) = n(n+1)$$

□

2 Geometric progressions

Definition 2: geometric progression

A sequence of real numbers is a **geometric progression** if the ratio between each term (starting from the second one) and the previous one is constant. This constant is called the common ratio of the progression.

Given the first term $a \in \mathbb{R}$ and the common ratio $q \in \mathbb{R}$, the associated geometric progression is:

$$a, a q, a q^2, a q^3, a q^4, \dots$$

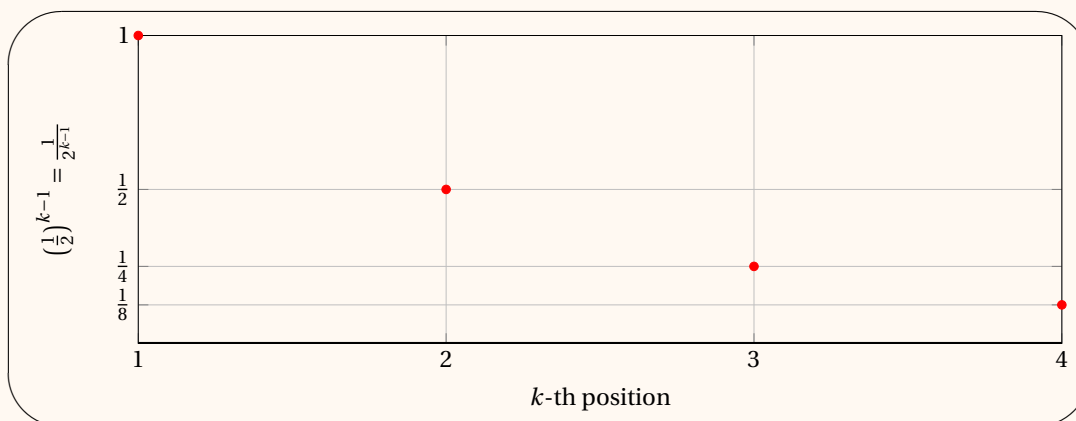
Each term (starting from the second one) is obtained from the previous one by multiplying it by q . The k -th term ($k \in \mathbb{N}$, $k \geq 1$) can be written as $a q^{k-1}$ and we have:

$a q^{1-1} = a q^0 = a$	first term, in position $k = 1$
$a q^{2-1} = a q^1 = a q$	second term, in position $k = 2$
$a q^{3-1} = a q^2$	third term, in position $k = 3$
$a q^{4-1} = a q^3$	fourth term, in position $k = 4$
$\dots$	$\dots$

Example 3: geometric progressions

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$$



- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, 2, 4, 8$$

2.1 Sums of the terms of geometric progressions

Remark 7: sum of the first n terms of the geometric progression ($a = 1$)

Given $q \in \mathbb{R}_+$, for every natural number $n \geq 1$ we have:

$$\sum_{k=1}^n q^{k-1} = \begin{cases} \frac{q^n - 1}{q - 1} & \text{if } q \neq 1 \\ n & \text{otherwise} \end{cases} \quad (7)$$

Proof. If $q \neq 1$, we prove the equation in the following equivalent form:

$$(q - 1) \sum_{k=1}^n q^{k-1} = q^n - 1$$

Applying the properties of summations, we get:

$$\begin{aligned} (q - 1) \sum_{k=1}^n q^{k-1} &= q \sum_{k=1}^n q^{k-1} - \sum_{k=1}^n q^{k-1} = \\ &= \sum_{k=1}^n q^k - \sum_{k=1}^n q^{k-1} = \sum_{k=1}^n q^k - \sum_{k=0}^{n-1} q^k = \\ &= \sum_{k=1}^{n-1} q^k + q^n - \left(1 + \sum_{k=1}^{n-1} q^k \right) = q^n - 1 \end{aligned}$$

If $q = 1$, we have instead:

$$\sum_{k=1}^n q^{k-1} = \sum_{k=1}^n 1 = n$$

□

- Given $q \in \mathbb{R}_+$, $n \in \mathbb{N}$, $n \geq 1$ and $a \in \mathbb{R}$, formula (7) extends as follows:

$$\sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{q^n - 1}{q - 1} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases} \quad \text{since } \sum_{k=1}^n a q^{k-1} = a \sum_{k=1}^n q^{k-1} \quad (8)$$

clearly:

$$\frac{q^n - 1}{q - 1} = \frac{1 - q^n}{1 - q} \quad \text{and hence we also have } \sum_{k=1}^n a q^{k-1} = \begin{cases} a \left(\frac{1 - q^n}{1 - q} \right) & \text{if } q \neq 1 \\ a n & \text{otherwise} \end{cases}$$

Example 4: sum of the first n terms of geometric progressions

- With $a = 1$ and $q = \frac{1}{2}$, the first 4 terms are:

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8} \quad \text{and} \quad 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{8+4+2+1}{8} = \frac{15}{8}$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 \left(\frac{1}{2}\right)^{k-1} = \sum_{k=1}^4 \frac{1}{2^{k-1}} = \frac{1 - \frac{1}{2^4}}{1 - \frac{1}{2}} = \frac{1 - \frac{1}{16}}{\frac{1}{2}} = \frac{15}{16} \cdot 2 = \frac{15}{8}$$

- With $a = 1$ and $q = 2$, the first 4 terms are:

$$1, 2, 4, 8 \quad \text{and} \quad 1 + 2 + 4 + 8 = 15$$

the sum of the first $n = 4$ terms is given by the formula:

$$\sum_{k=1}^4 2^{k-1} = \frac{2^4 - 1}{2 - 1} = 16 - 1 = 15$$